

1. AI Tool(s) Used

- ChatGPT
- Gemini
- Claude

2. Scope of Usage

I used AI tools to refactor almost all of my code in this project. Initially, I built the chatroom using React, but I needed to convert it to pure Vanilla JavaScript, HTML, and CSS.

A. Refactoring React to Vanilla JS

Location: Entire project (index.html, style.css, js files)

Prompt: "I have a React chatroom app and I need to convert it entirely to Vanilla JS without any build tools. Please help me refactor."

Response: The AI generated the equivalent Vanilla JS code, mapping my React components to standard DOM manipulation and HTML elements.

Refinement: I had to go through the generated code to fix several bugs, especially with event listeners not attaching correctly and the CSS layout breaking. I manually adjusted the CSS flexbox properties to fix the scrolling issues.

B. Firebase Integration

Location: js/chat.js and js/auth.js

Prompt: "How to rewrite my React Firebase hooks to work in Vanilla JS using the compat SDK?"

Response: AI provided the onSnapshot functions to listen to the Firestore collections and auth functions.

Refinement: I modified the logic to properly handle my specific UI requirements, like filtering out blocked users and updating the chat list dynamically.

C. Gemini Bot Integration

Location: js/chat.js

Prompt: "How to use the Gemini API in a basic JavaScript fetch request?"

Response: AI provided the endpoint and the JSON structure.

Refinement: I wrapped the fetch call inside my chat logic and added a trigger so it only responds when someone types "@bot".

3. Statement of Non-Usage

While AI did the heavy lifting for the refactoring process, the original logic, design, and features were planned and developed by me in the initial React version. I manually verified and fixed the AI's vanilla JS output to ensure it worked perfectly.